

HDOC Day-12 Assignment

Aim : write a menu-driven c code using switch-case to input a positive integer and then perform specific operations on its digits.

1. ALGORITHM

- Start the program.
- Display the menu options to user.
 - Op 1 : find the sum and product of the first and last digits of a given number.
 - Option 2 : count digits lesser than, equal to and greater than a given digit 'k'
 - Option 3 ; exit.
- Read the user's selected choice.
- Using a switch-case statement .
- **If Choice == 1:**
 - Input a positive integer num.
 - Find the last digit using `last_digit = num % 10`.
 - Copy num to a temporary variable temp.
 - Find the first digit using a while loop (`temp >= 10`) by dividing temp by 10 continuously.
 - Calculate `sum = first_digit + last_digit`.
 - Calculate `product = first_digit * last_digit`.
 - Print sum and product.
- **If Choice == 2:**
 - Input a positive integer num and a single digit k.
 - Initialize variables `count_lesser = 0`, `count_equal = 0`, and `count_greater = 0`.
 - Copy num to temp.
 - Use a while loop (`temp > 0`) to extract each digit (`digit = temp % 10`):
 - If `digit < k`, increment `count_lesser`.
 - If `digit == k`, increment `count_equal`.
 - If `digit > k`, increment `count_greater`.
 - Remove the last digit from temp (`temp = temp / 10`).
 - Print `count_lesser`, `count_equal`, and `count_greater`.

- **If Choice == 3:**
 - Print "Exiting program! " and stop execution.
- **Default Case:**
 - Print "Invalid choice, try again."
- Stop the program.

- **TEST CASE TABLE :**

TEST CASE	CHOICE	INPUT (num)	Target digit 'k'	Expected op	status
1.	1	9824	na	Sum = 13, Product = 36	pass
2.	2	48294	4	Count Lesser = 1, Count Equal = 2, Count Greater = 2	pass
3.	3	na	na	Exit program !	pass

- **C Program code:**

```
UW PICU 5.09

#include <stdio.h>

int main() {
    int choice;
    int num, temp;
    int first_digit, last_digit;
    int sum, product;
    int k, digit;
    int count_lesser = 0, count_equal = 0, count_greater = 0;

    printf("--- MENU ---\n");
    printf("1. Sum and Product of First and Last Digits\n");
    printf("2. Count Digits Lesser, Equal, or Greater than 'k'\n");
    printf("3. Exit\n");
    printf("Enter your choice: ");
    scanf("%d", &choice);

    switch(choice) {
        case 1:
            printf("Enter a positive integer: ");
            scanf("%d", &num);

            last_digit = num % 10;
            temp = num;

            while(temp >= 10) {
                temp = temp / 10;
            }
            first_digit = temp;

            sum = first_digit + last_digit;
            product = first_digit * last_digit;

            printf("Sum = %d\n", sum);
            printf("Product = %d\n", product);
            break;
```

```

case 2:
    printf("Enter a positive integer: ");
    scanf("%d", &num);
    printf("Enter the digit k (0-9): ");
    scanf("%d", &k);

    temp = num;

    while(temp > 0) {
        digit = temp % 10;

        if(digit < k) {
            count_lesser++;
        }
        else if(digit == k) {
            count_equal++;
        }
        else {
            count_greater++;
        }

        temp = temp / 10;
    }

    printf("Count_lesser = %d\n", count_lesser);
    printf("Count_equal = %d\n", count_equal);
    printf("Count_greater = %d\n", count_greater);
    break;

case 3:
    printf("Exiting program...\n");
    break;

default:
    printf("Invalid choice!\n");
}

return 0;
}

```

- **Program Execution Output**
*** for option 1 and 2 .**

```
[manalithakar@Unknown_8e:80:9d:9c:62:11 ~ % gcc 12q1.c -o 12q1
[manalithakar@Unknown_8e:80:9d:9c:62:11 ~ % ./12q1
--- MENU ---
1. Sum and Product of First and Last Digits
2. Count Digits Lesser, Equal, or Greater than 'k'
3. Exit
Enter your choice: 1
Enter a positive integer: 9824
Sum = 13
Product = 36
[manalithakar@Unknown_8e:80:9d:9c:62:11 ~ % ./12q1
--- MENU ---
1. Sum and Product of First and Last Digits
2. Count Digits Lesser, Equal, or Greater than 'k'
3. Exit
Enter your choice: 2
Enter a positive integer: 48294
Enter the digit k (0-9): 4
Count_lesser = 1
Count_equal = 2
Count_greater = 2
[manalithakar@Unknown_8e:80:9d:9c:62:11 ~ % ./12q1
--- MENU ---
1. Sum and Product of First and Last Digits
2. Count Digits Lesser, Equal, or Greater than 'k'
3. Exit
Enter your choice: 3
Exiting program...
manalithakar@Unknown_8e:80:9d:9c:62:11 ~ %
```